

INTERVAL TIMER

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Often, we forget to switch off some of the electrical loads/devices in our houses/offices after their usage, especially those with short duration requirements such as bathroom/toilet exhaust fans and lights. This leads to wastage of electricity and increased bills.

To overcome this problem, we present a simple interval timer circuit. When this circuit is connected between the power supply and the load, it automatically switches off power supply to the load after a

preset time (in this case 15 minutes). This ensures that the devices are used only for the necessary amount of time, resulting in energy savings.

To switch on the device again, one must switch off the circuit and then switch it on to get another preset time usage. Fig. 1 shows the prototype tested in EFY lab.

Circuit and working

Fig. 2 shows the circuit diagram of interval timer. It is built around a 230V AC primary to 12V, 1A secondary

transformer (X1), five 1N4007 rectifier diodes (D1 through D5), CD4541BE (IC1), 12V voltage regulator LM7812 (IC2), npn relay driver transistor 2N2219 (T1), a 12V one changeover relay

(RL1), and a few other components.

An interval timer is a timer that turns on its output as soon as input switch (not shown in circuit diagram) is turned on and 230V AC is available across its input (CON1 in Fig. 2). Its output will turn off after a set time and will remain even if 230V AC input is still available across CON1. The timer circuit described here is based on CMOS IC CD4541BE (IC1). This IC has a built-in oscillator and binary counters used as frequency divider to achieve long time delays. The oscillator frequency (f) is decided by external RC components, where $f = 1/(2.3 \times R2 \times C1)$.

The oscillator frequency is reduced by the programmable binary counter, which operates as a 2^N frequency divider. With the connection of IC1's pin nos. 12 and 13 to either ground (0V) or 12V, the frequency division factor 2^N can be changed as per the following table.

Pin 12	Pin 13	2^N	Count
0	0	2^{13}	8192
0	1	2^{10}	1024
1	0	2^8	256
1	1	2^{16}	65536

To achieve a time delay of approximately 15 minutes, the oscillator frequency is set to 35.6Hz, and the divide by 2^N counter is set to $2^{16} = 65536$. The chip has other features like single cycle/continuous on/off cyclic mode selection (pin 10), output initial selection after reset as either 0 or 1 (pin 9), auto reset on/off (pin 5) and master reset on/off (pin 6). Please refer the IC data sheet for complete details.

Functioning of the timer

The timer circuit is powered by 12V

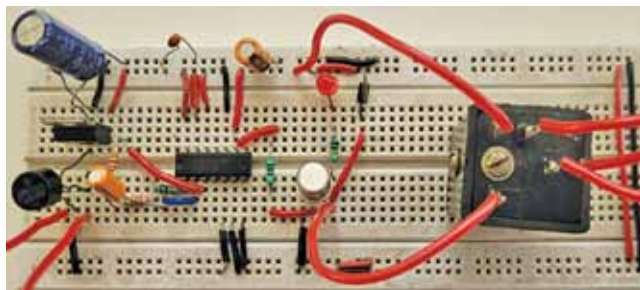


Fig. 1: Prototype tested in EFY Lab

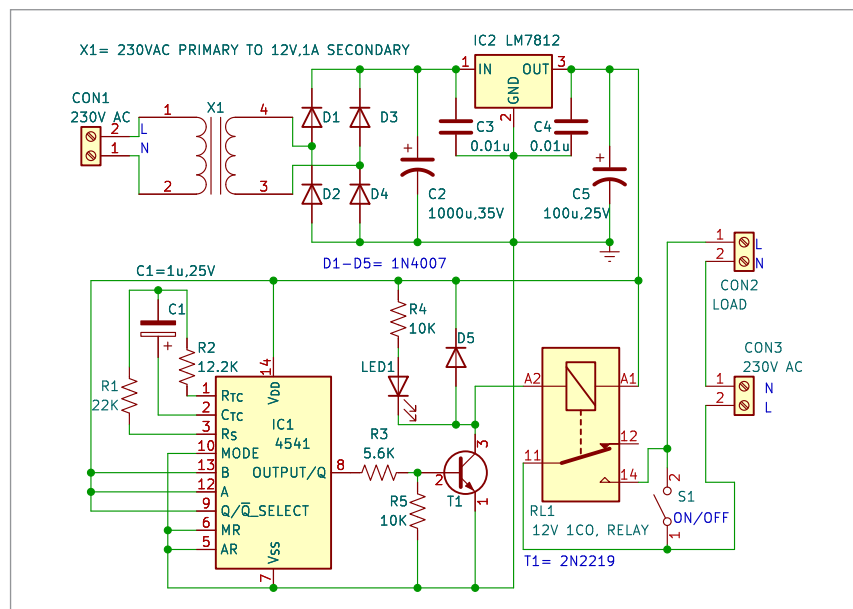


Fig. 2: Circuit diagram of interval timer